

EE 446 - HW 2

August 4, 2026

1 EE 446 – TinyML – Homework 2 Writeup

1.0.1 Ben Park

1.1 Question 3(c) — Base Model Evaluation

Classification Report and Confusion Matrix for `base_model` on the test set (`X_test`).

```
Classification Report (base_model):
              precision    recall  f1-score   support

     0       1.00      1.00      1.00     11702
     1       1.00      1.00      1.00     13493

 accuracy          1.00          1.00          1.00     25195
 macro avg          1.00          1.00          1.00     25195
 weighted avg          1.00          1.00          1.00     25195

Confusion Matrix (base_model):
[[11686   16]
 [   18 13475]]
```

1.2 Question 4(b) — Quantized Model Evaluation

Classification Report and Confusion Matrix for `tfllite_quant_model` on the test set (`X_test`), using properly quantized input samples via the input tensor's scale and zero point.

```
Classification Report (tfllite_quant_model):
              precision    recall  f1-score   support

     0       1.00      0.99      0.99     11702
     1       0.99      1.00      1.00     13493

 accuracy          1.00          1.00          1.00     25195
 macro avg          1.00          1.00          1.00     25195
 weighted avg          1.00          1.00          1.00     25195

Confusion Matrix (tfllite_quant_model):
[[11608    94]
 [   24 13469]]
```

1.3 Question 5(c) — On-Device Inference: First 5 Samples

Serial Monitor output from the Arduino Nano 33 BLE for the initial 5 test samples (Sample #, Predicted Class, Actual Class).

```
Sample #: 3 Predicted Class: 0 Actual Class: 0
Sample #: 4 Predicted Class: 1 Actual Class: 1
Sample #: 0 Predicted Class: 1 Actual Class: 1
Sample #: 1 Predicted Class: 1 Actual Class: 1
Sample #: 2 Predicted Class: 0 Actual Class: 0
```

1.4 Question 5(d) — On-Device Inference: Next 10 Samples

Serial Monitor output from the Arduino Nano 33 BLE for the additional 10 test samples (excluding the first 5), confirming deployment on new data.

```
Sample #: 0 Predicted Class: 1 Actual Class: 1
Sample #: 1 Predicted Class: 1 Actual Class: 1
Sample #: 2 Predicted Class: 1 Actual Class: 1
Sample #: 3 Predicted Class: 0 Actual Class: 0
Sample #: 4 Predicted Class: 0 Actual Class: 0
Sample #: 5 Predicted Class: 0 Actual Class: 0
Sample #: 6 Predicted Class: 1 Actual Class: 1
Sample #: 7 Predicted Class: 1 Actual Class: 1
Sample #: 8 Predicted Class: 1 Actual Class: 1
Sample #: 9 Predicted Class: 1 Actual Class: 1
```